

Sitka, Alaska

Temperature Analysis — 2021

A full-year review of daily high and low temperatures in Sitka, Alaska, based on recorded weather data for 2021. This report examines seasonal patterns, temperature extremes, and the relationship between daily highs and lows across the year.

Overview

Sitka is a coastal city in Southeast Alaska, situated on Baranof Island along the Pacific Ocean. Its climate is classified as oceanic — characterized by mild, wet winters and cool summers. The Pacific moderates temperature extremes year-round, making Sitka significantly warmer in winter than interior Alaskan cities, while keeping summers cooler than most of the continental US.

The dataset covers 365 days of 2021, tracking both the daily high and daily low temperatures. All temperatures are recorded in degrees Fahrenheit (°F). Visualizations were generated using Python with matplotlib.

Daily High Temperatures

The daily high temperatures show a smooth and well-defined seasonal curve over 2021. The year opened with highs in the low-to-mid 40s°F, consistent with Sitka's mild oceanic winters. Temperatures held relatively steady through February and March before beginning a gradual climb into spring.

A notable cold anomaly occurred in late February, where the daily high briefly dropped to near 18°F — a sharp departure from the seasonal trend, likely driven by an arctic air intrusion pushing cold continental air into the coastal region.

From May onward, highs rose steadily through the summer months. The peak of the year was recorded in mid-July at approximately 78°F. The summer plateau between June and August showed consistent highs in the 55–75°F range, with notable day-to-day variability. The autumn descent began in September, and by December, highs had fallen back into the mid-to-upper 20s°F.

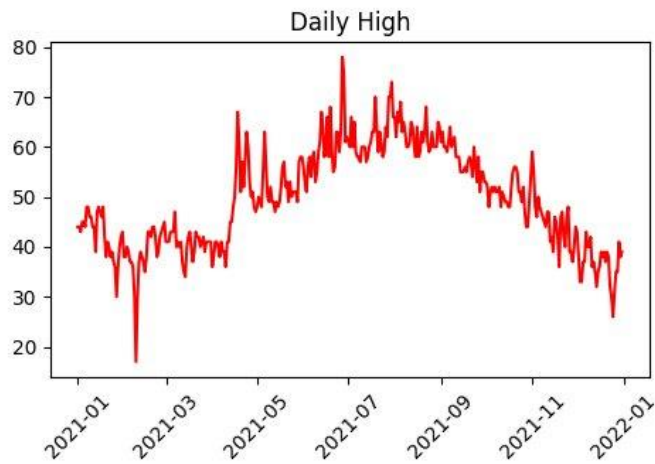


Figure 1: Daily High Temperatures, Sitka 2021

Daily Low Temperatures

The daily low temperatures followed a broadly similar seasonal arc to the highs, but with more pronounced winter extremes. January opened with lows in the upper 30s°F, before a dramatic plunge in late February brought lows down to approximately 12°F — the coldest recorded value of the entire year.

Spring recovery was gradual but consistent. By April and May, lows had climbed back into the 30s and 40s°F. The summer months of July and August saw the warmest overnight temperatures, with lows peaking near 58–60°F — reflecting the moderating effect of long daylight hours and warm sea surface temperatures.

Autumn brought a steady but uneven decline. Lows dropped sharply in November and December, finishing the year near 15°F by early January 2022 — mirroring the late February anomaly and suggesting that 2021 was bookended by two unusually cold periods for Sitka's standards.

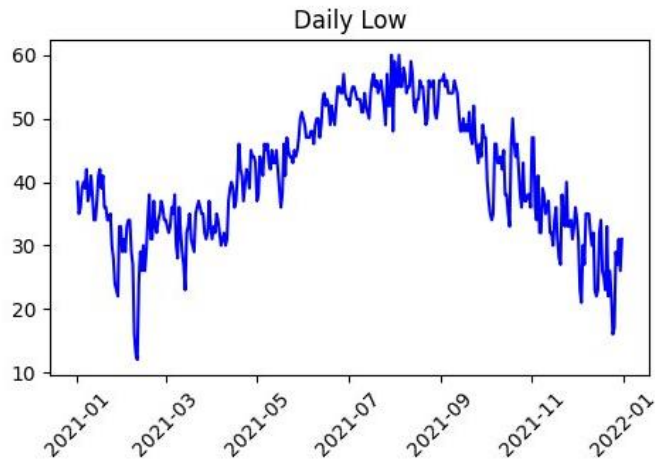


Figure 2: Daily Low Temperatures, Sitka 2021

Combined High vs Low Comparison

Plotting the daily highs and lows together reveals important patterns about the diurnal temperature range — the difference between the warmest and coolest part of each day.

During winter months, the gap between highs and lows is relatively wide. In summer, the two lines converge noticeably: long Alaskan days and cloudy maritime conditions keep overnight temperatures from dropping far below afternoon peaks, resulting in a narrower daily range.

The most dramatic divergence occurs around the late February cold event, where the low dropped to 12°F while the high only reached the upper teens — a near-freezing day that stands out clearly against the rest of the year's trend. Overall, the chart reinforces Sitka's oceanic character: a climate where extreme heat and extreme cold are both rare.

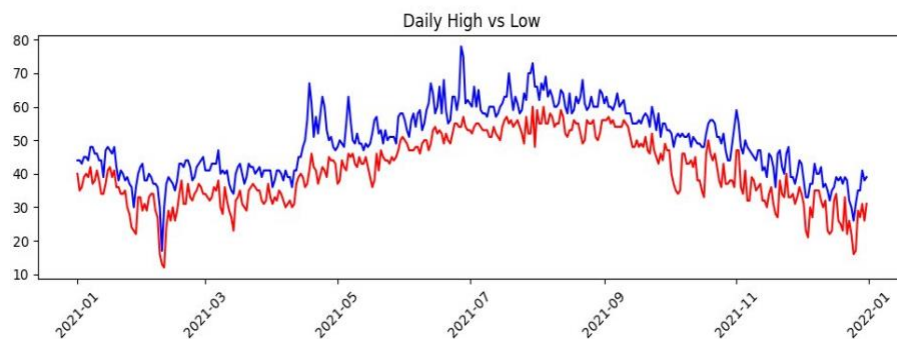


Figure 3: Daily High vs Low Temperatures Overlaid, Sitka 2021

Key Takeaways

- Warmest day: ~78°F high in mid-July 2021
- Coldest low: ~12°F in late February — an anomalous arctic intrusion
- Summer highs (June–August): consistently 55–75°F
- Summer lows (July–August): peaked near 58–60°F, narrowing the diurnal range
- Annual high range: ~27°F (Dec) to ~78°F (Jul) — a 51°F spread
- Annual low range: ~12°F (Feb) to ~60°F (Aug) — a 48°F spread
- Climate character: Classic oceanic — mild, maritime, with rare temperature extremes

Conclusion

Sitka's 2021 temperature record paints a picture of a stable, maritime climate shaped by its coastal geography. The year followed a predictable seasonal arc, punctuated by a notable cold anomaly in late February. Summers were mild and pleasant, winters cool but rarely severe. The narrow diurnal range in summer and the moderating influence of the Pacific Ocean throughout the year are hallmarks of Sitka's climate — distinguishing it sharply from the more extreme conditions found in Alaska's interior.